Marketing Analytics

Visualization and Data Quality

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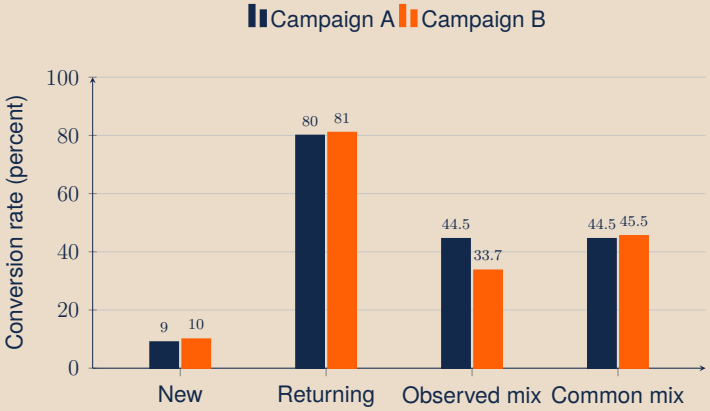
Session 03

A Campaign Comparison

- Campaign B converts more new and returning customers within each group.
- Its aggregate conversion rate is lower.
- The customer mix determines which comparison the aggregate makes.

Customer Mix Reverses the Ranking



Synthetic campaigns. Both within-group rates rise by one percentage point.

A Standardized Conversion Rate

Use common customer-group weights w_g for each campaign c.

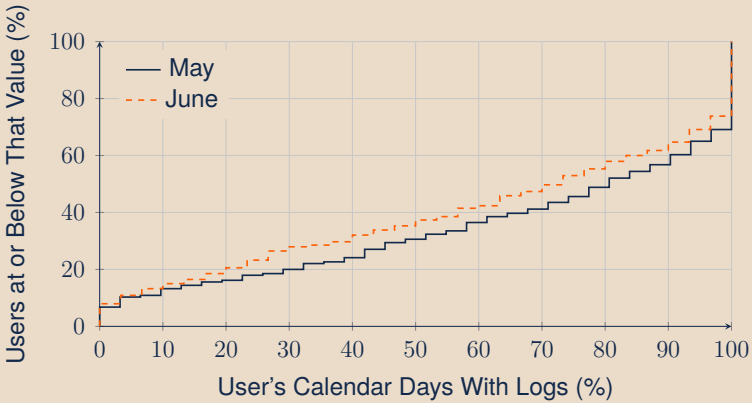
$$\bar{p}_c^{\text{std}} = \sum_g w_g p_{cg}, \qquad \sum_g w_g = 1$$

Changing the weights changes the population represented by the rate.

Chart Choice Follows the Question

- Aligned positions compare magnitudes.
- Lines compare changes over ordered time.
- Distributions reveal spread and concentration.
- Paired displays preserve within-customer changes.

A Mean Decline Can Hide a Stable Median



mHealth comparison uses paired customers and a common observation window.

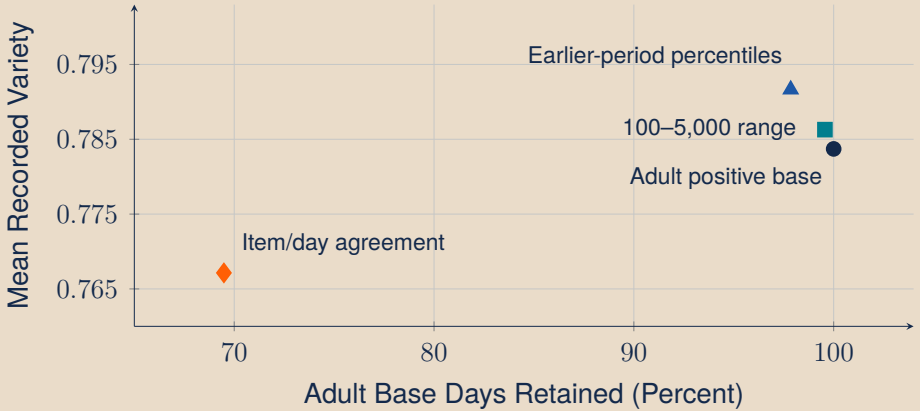
Uncertainty and Dependence

- Repeated days for one customer share common influences.
- Customer-level resampling preserves each customer history.
- A confidence interval and a percentile range describe different objects.

Nutrition Knowledge and Measurement

- A recorded food log can omit meals and portions.
- Energy plausibility depends on the measurement instrument and physiology.
- Age, body size, activity, and recording duration affect screening.

Sample Restrictions Change the Comparison



mHealth screening alternatives are explicit analysis choices.

Energy-Plausibility Screens

- An intake-to-requirement comparison needs defined inputs.
- Unobserved physiological inputs remain missing.
- A hypothetical Goldberg calculation teaches the method separately from the logs.

Interactive Comparisons

- Keep comparable panels on common scales.
- Display the eligible population and current selection.
- Preserve customers with no recorded days when the estimand includes them.

A Composition Check

Synthetic within-group rates are 9%, 80%, 10%, and 81%.

```
rates_a = [0.09, 0.80]
rates_b = [0.10, 0.81]
common_a = sum(rates_a) / 2
common_b = sum(rates_b) / 2
observed_b = (2 * rates_b[0] + rates_b[1]) / 3
assert round(common_b - common_a, 2) == 0.01
```

B improves by one point under common weights despite its lower observed aggregate.

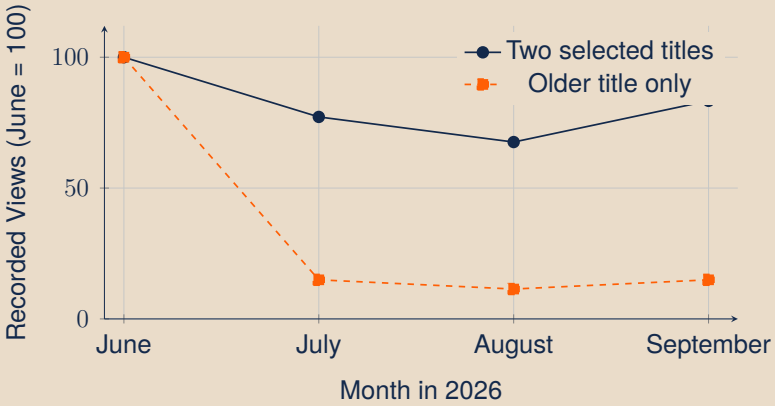
Coding Exercise: A Chart Audit

- Generate a rate chart from a verified aggregate table.
- Alter the axis range, cohort filter, and denominator one at a time.
- Explain which changes alter the substantive comparison.

A Public-Signal Measurement Problem

- A product can acquire a new page title while demand stays unchanged.
- A single-title series then misses part of its recorded attention.
- Coverage, redirects, and missing dates require an audit.

Title Changes and Apparent Declines



The supplied public pageview capture uses observed title histories.

Evidence in a Management Presentation

- State the decision and supported result.
- Show the comparison, units, and denominator.
- Keep the qualification that could change the action beside the claim.

Honest Comparisons

- Keep denominators and scales visible.
- Preserve missing outcomes and relevant subgroup variation.
- Avoid presenting a filtered descriptive pattern as a general causal effect.

Visualization Discussion

- Which customers receive weight in the chart?
- Could sample restrictions reverse the recommendation?
- What does the uncertainty interval include?